

ensemble model & ensemble of small model

“All models are wrong, some are useful”

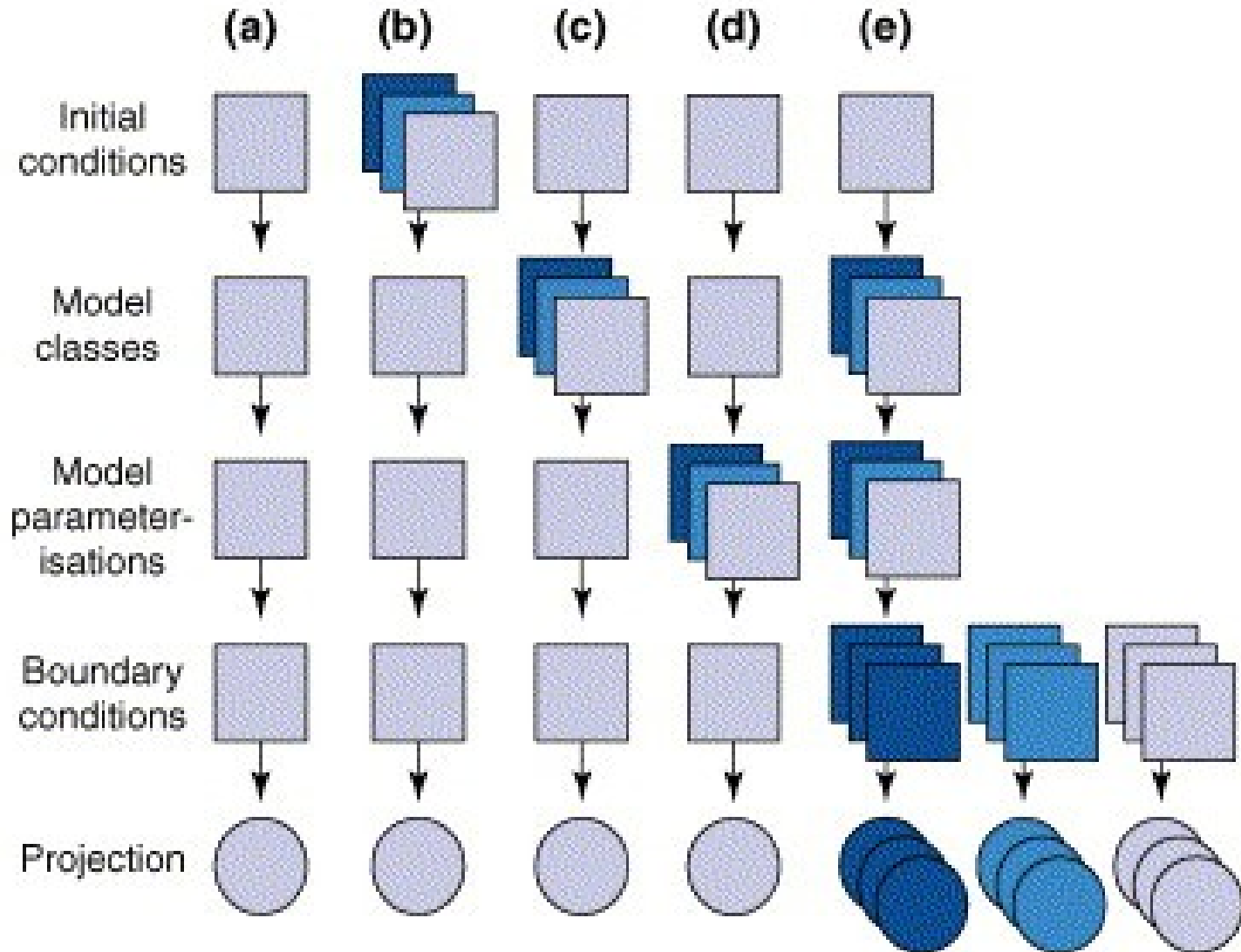
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- In 1976, a British statistician named George Box wrote the famous line, “**All models are wrong, some are useful.**”
- “**Nothing in Biology Makes Sense Except in the Light of Evolution**” is a 1973 essay by the evolutionary biologist Thodosius Dobzhansky;

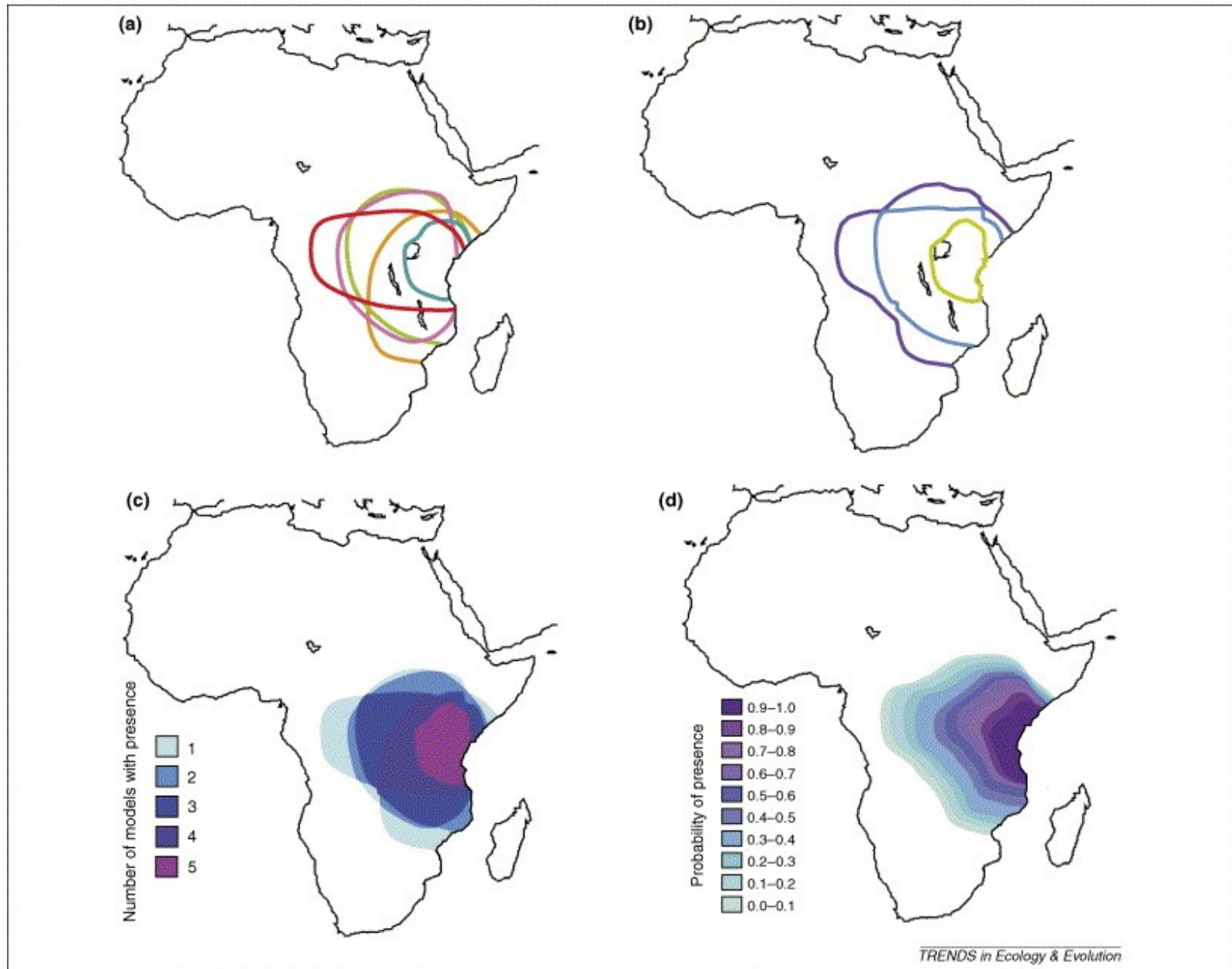
Occ data → Model strategy

- ❖ For species that have 1-10 points: No model, try ecoregion (ecoregion also can be used to delimit model calibration area).
- ❖ For species that have > 20 points: ensemble model (I will show this with sdm).
- ❖ For species that have 10-20 points: ensemble of small model (I will show this with FlexSDM)

Why we need ensemble model ?

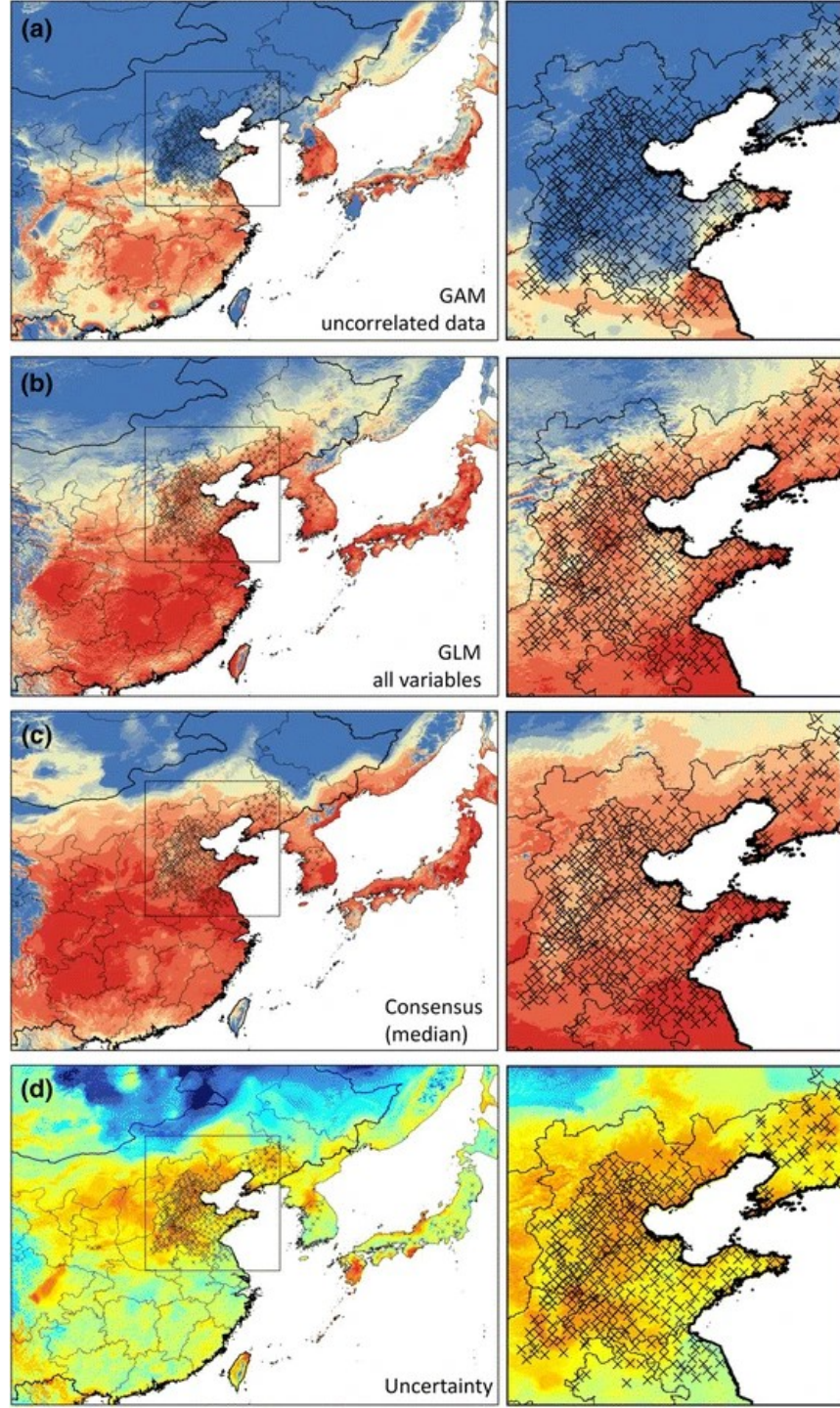


Why ensemble model ?

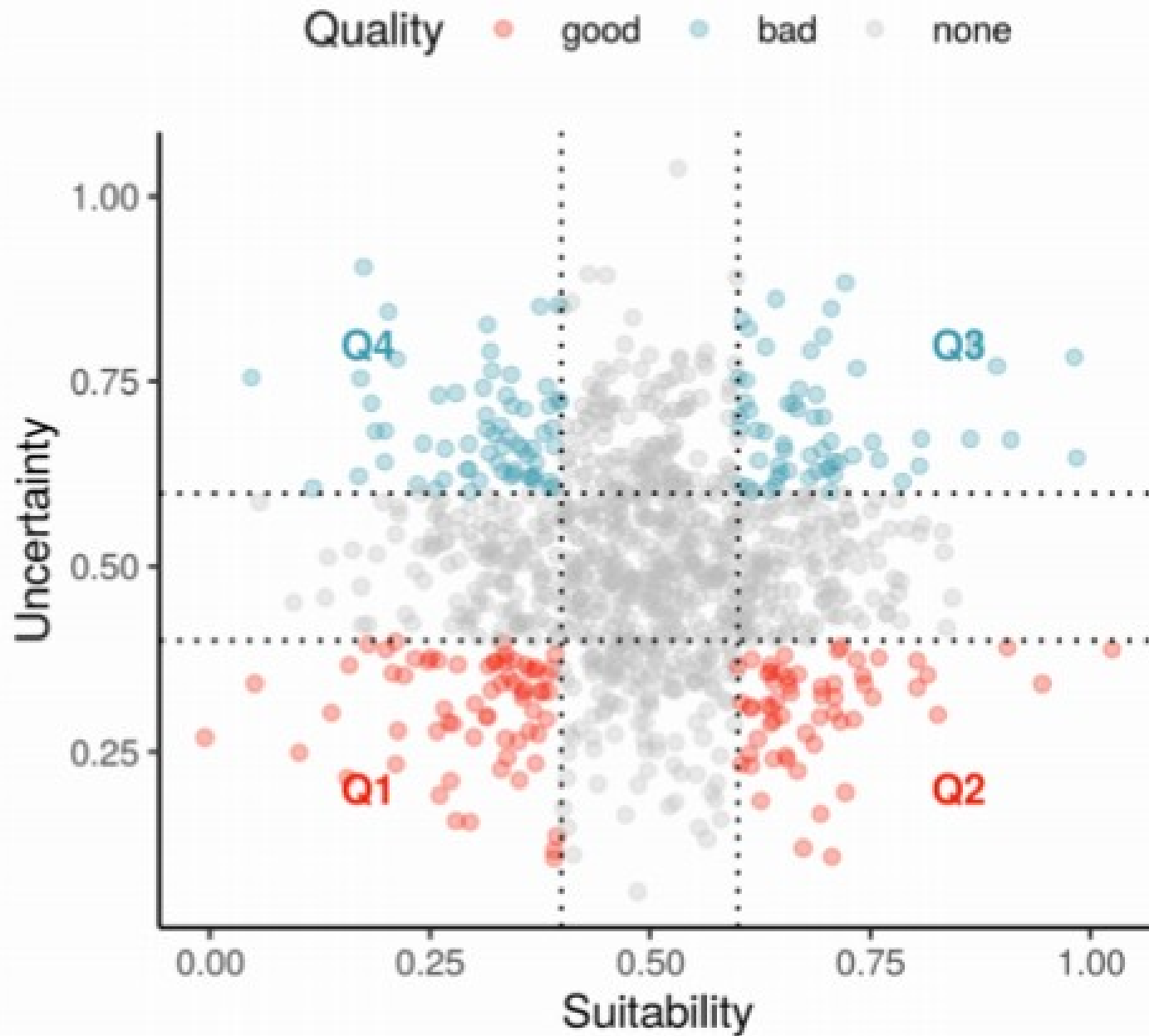


Individual and consensus models predictions of Fall Webworm in East Asia.

Zhu & Peterson 2017

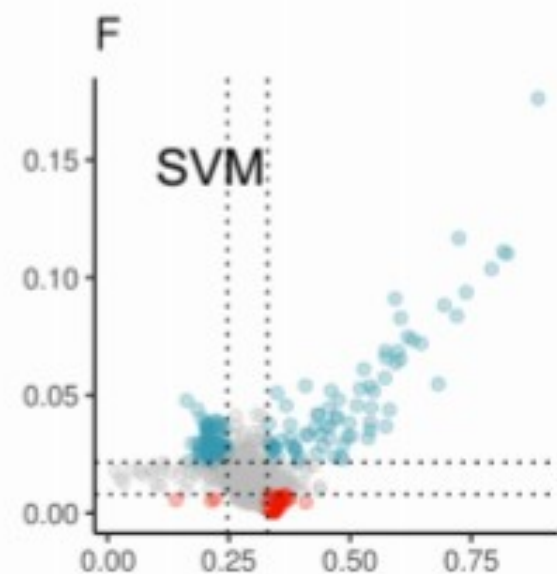
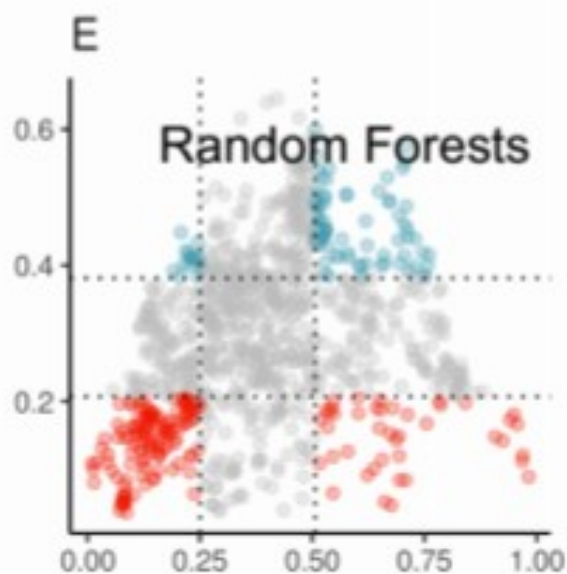
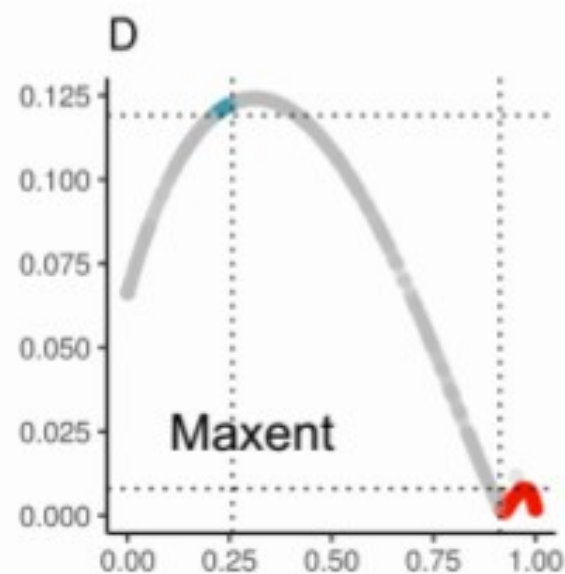
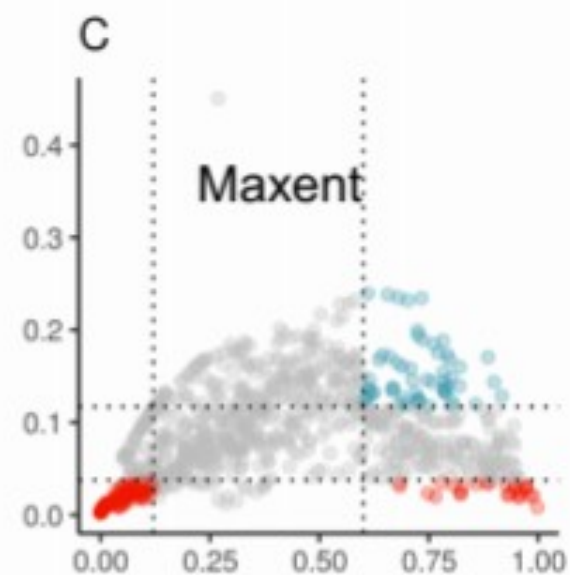
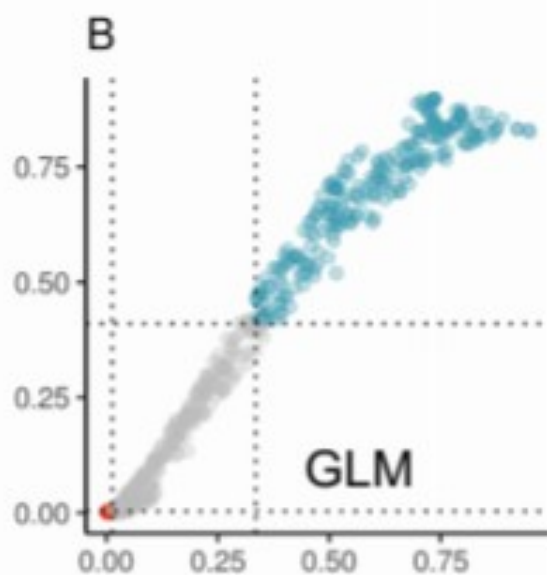
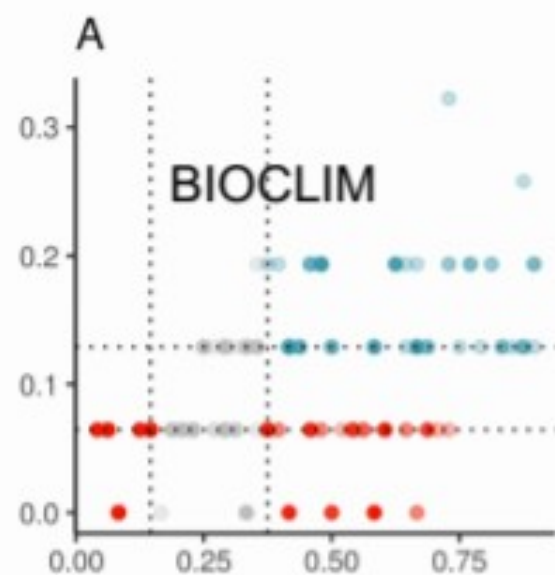


Why ensemble model?

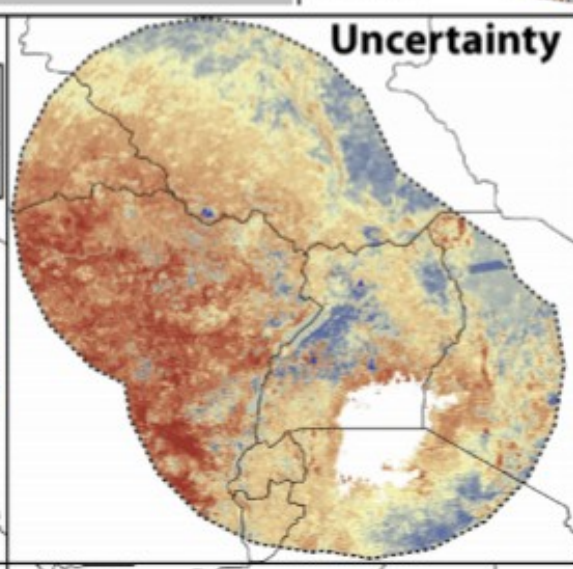
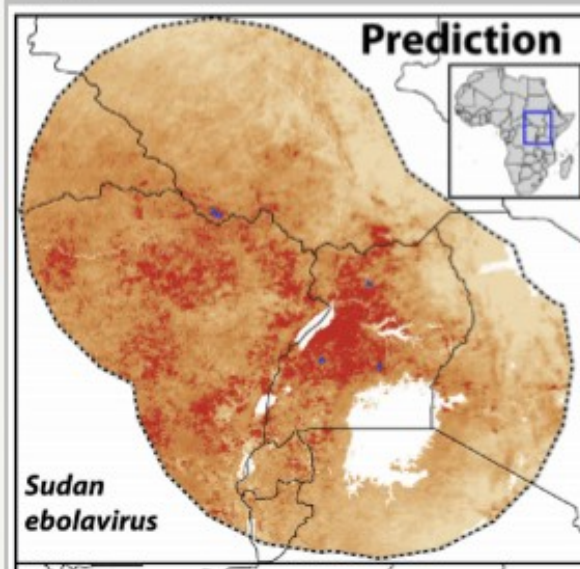
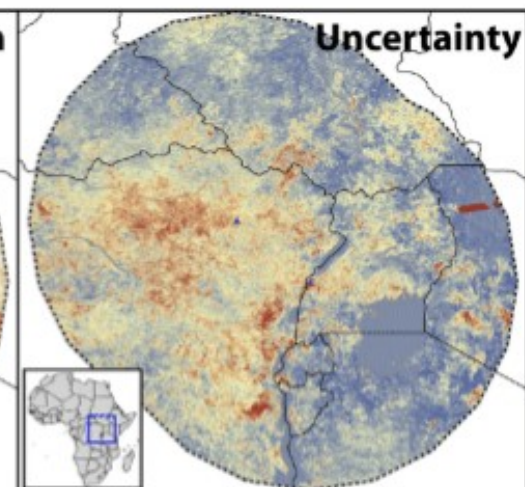
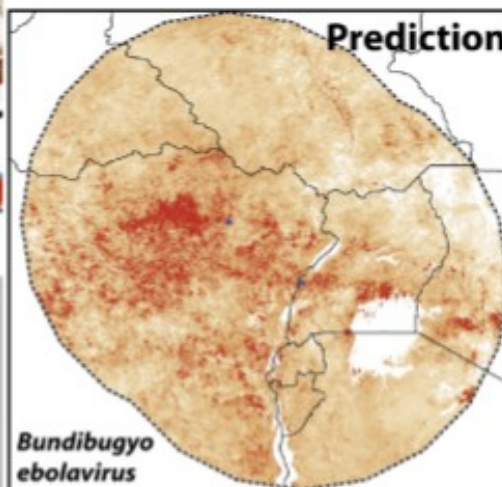


Quality ● good ● bad ● none

Uncertainty



Suitability



Ecological niche modeling and potential dispersal of emerald ash borer in the Pacific Northwest

Gengping Zhu ✉, Max Ragozzino, Mark Cody Holthouse, Matthew Mills, Jessica L Celis, Stacy Johnson, David W Crowder

Journal of Economic Entomology, Volume 118, Issue 5, October 2025, Pages 2404–2411,
<https://doi.org/10.1093/jee/toaf175>

Published: 15 July 2025 Article history ▼

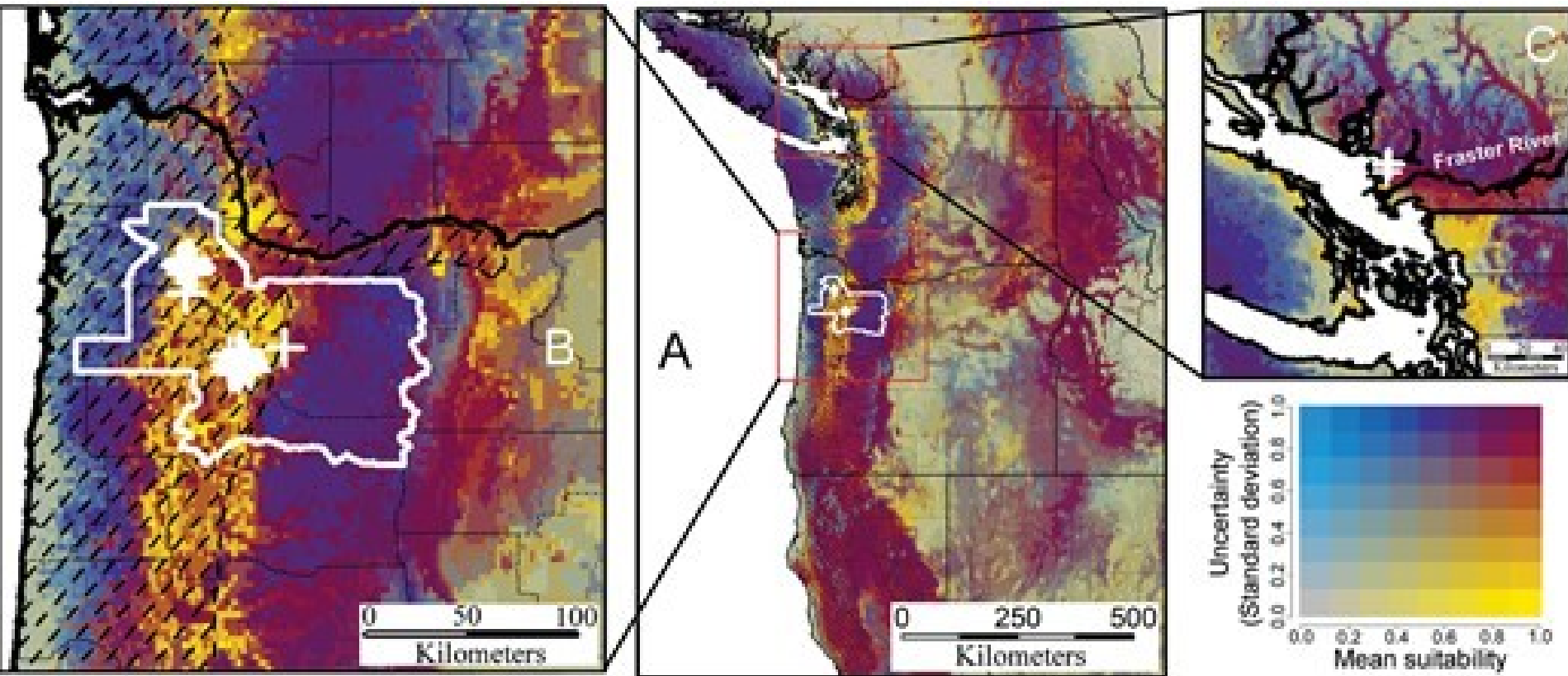


Fig. 3. Ensemble habitat suitability model prediction for emerald ash borer in the Pacific Northwest. Bivariate maps denote ensemble predictions and their uncertainty; the increasing intensities of yellow represent increasing habitat suitability, and increasing blue represent increasing uncertainty.

Where these variations came from?

Methods in Ecology and Evolution



Standard Paper |  Free Access

Measuring the relative effect of factors affecting species distribution model predictions

Emeric Thibaud , Blaise Petitpierre, Olivier Broennimann, Anthony C. Davison, Antoine Guisan

3. We illustrate the framework by examining the relative importance of five key factors – a missing covariate, spatial autocorrelation due to a dispersal process in presences/absences, sample size, sampling design and modelling technique – in a real study framework based on virtual plants in a mountain landscape at regional scale, and show that, for the parameter values considered here, **most of the variation in prediction accuracy is due to sample size and modelling technique**. Contrary to repeatedly reported concerns, spatial autocorrelation has only comparatively small effects.

How to produce ensemble model?

- Ensemble model should be fitted based on individual models...
- These individual models could be fitted separately, by unique package,
- then, a consensus model is produced, i.e., mean, median, weighted average..

....

Generalized linear model (GLM) parameters...

- We can fit linear, quadratic or higher polynomial terms (check **poly()**) and interactions between predictors:
- the term **I()** indicates that a variable should be transformed before being used as predictor in the formula;
- **poly(x, n)** creates a polynomial of degree;
- **x1:x2** creates a two-way interaction term between variables x1 and x2, the linear terms of x1 and x2 would have to be specified separately;
- **x1*x2** creates a two-way interaction term between variables x1 and x2 **plus** their linear terms, i.e., **x1*x2 == 1x + x2 + x1:x2**
- **x1*x2*x3** creates the linear terms of the three variables, all possible two-way interactions between these variables and the three-way interaction

...

Generalized Additive Model (GAM)

parameters...

package: gam, mgcv

- used when the linear predictor depends linearly on unknown smooth functions of some predictor variables. The distinction is blurry as you can represent numeric covariables e.g. by a spline also in a GLM
- an adaptation that allows us to model **non-linear data** while maintaining explainability.
- allows to either shrink the importance of variables depending on the significance, or it allows to shrink insignificant variables to 'zero', meaning that this represents a variable selection.
- spline smoother $s()$ is related to model complexity, with the complexity parameter k set to 3 for a simple shape, and $k=-1$ for letting mgcv optimizing the complexity.
-

Random Forest (RF) parameters...

package: randomforest

- Machine learning method...
- use a bagging procedure for averaging the outputs of many different CARTs (classification and regression trees)
- CARTs bootstrapped samples of the training data and then either average the results in case of regression trees or make a simple vote in case of classification trees (committee averaging)
- An important feature of random forests are the out-of-bag samples, which means that the prediction/fit for a specific data point is only derived from averaging trees that did not include this data point during tree growing.
- **The output of Random Forests is essentially cross-validated!**
- Random forests estimate variable importance by a permutation procedure, which measures for each variable the drop in mean accuracy when this variable is permuted.

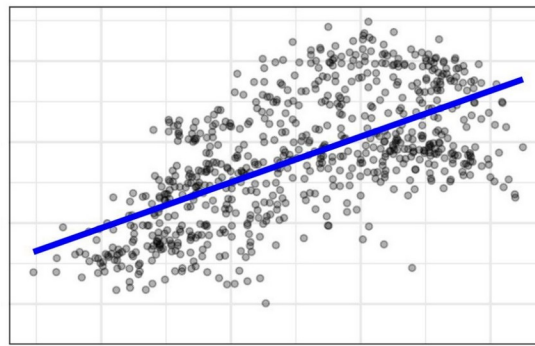
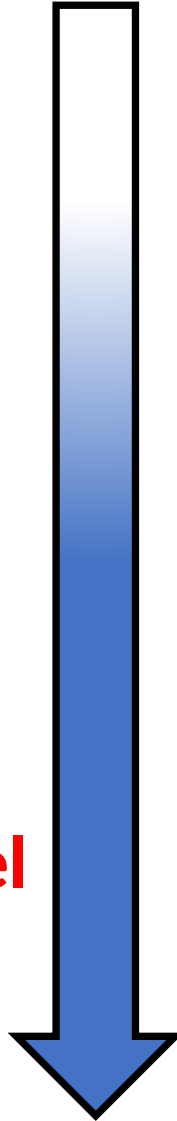
Maxent: maxnet package

- maxnet: Fitting 'Maxent' Species Distribution Models with 'glmnet'
- <https://cran.r-project.org/web/packages/maxnet/index.html>
- Procedures to fit species distributions models from occurrence records and environmental variables, using 'glmnet' for model fitting.
- Model structure is the same as for the 'Maxent' Java package, version 3.4.0, with the same feature types and regularization options.

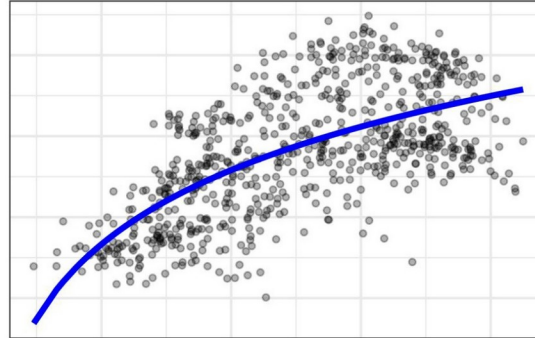
Simple model

Complexity

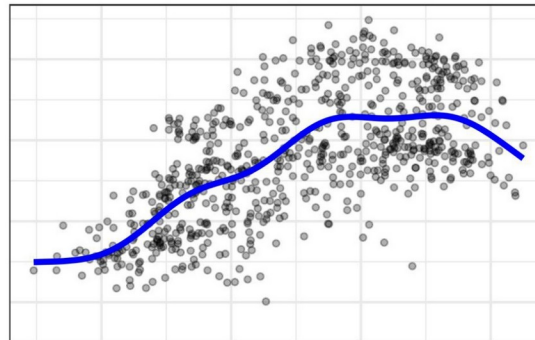
Complex model



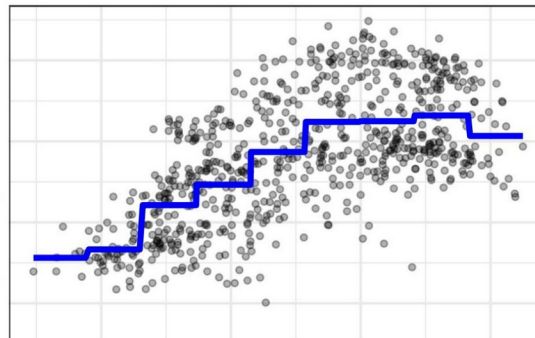
Linear model



Generalized linear model



Generalized additive model



**Machine learning
Random forest
Boot regression
Maxent**

How to produce ensemble model?

- Ensemble model should be fitted based on individual models...
- These individual models could be fitted separately, by unique package, then produce a consensus model;
- Now, these individual models can be fitted on single platform, example include:

dismo, Bimod2, sdm, flexsdm

dismo package

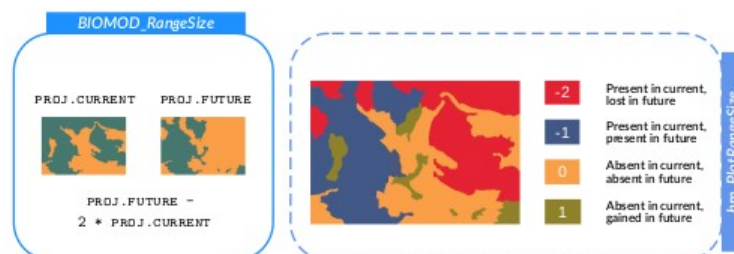
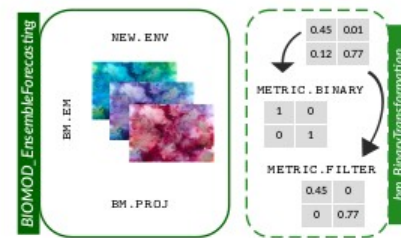
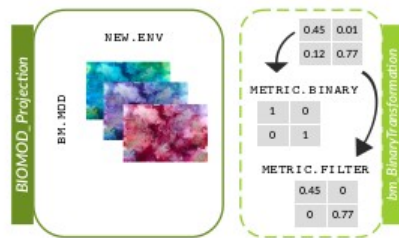
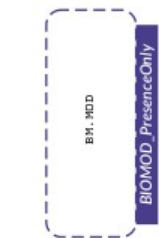
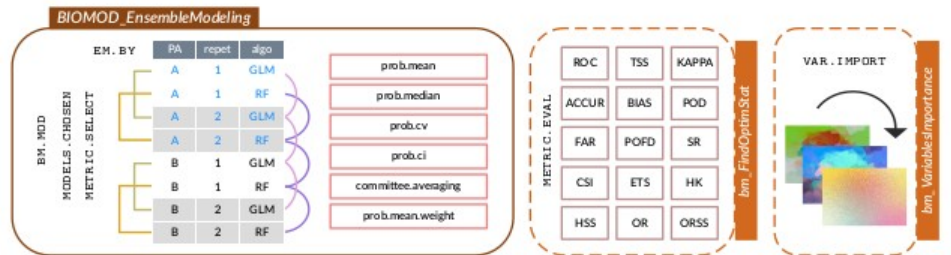
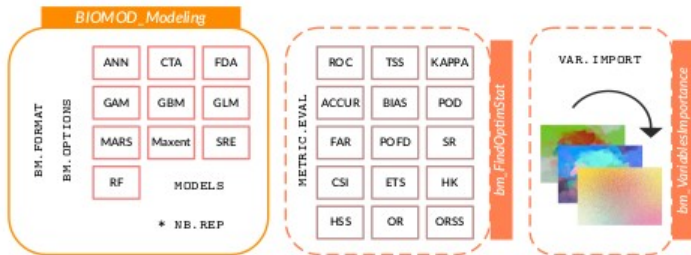
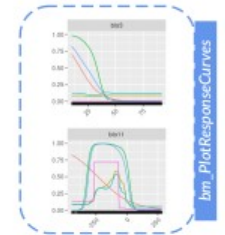
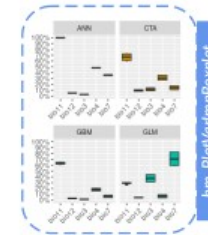
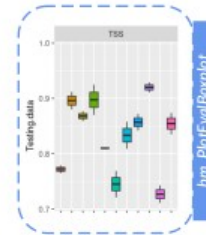
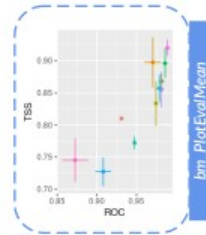
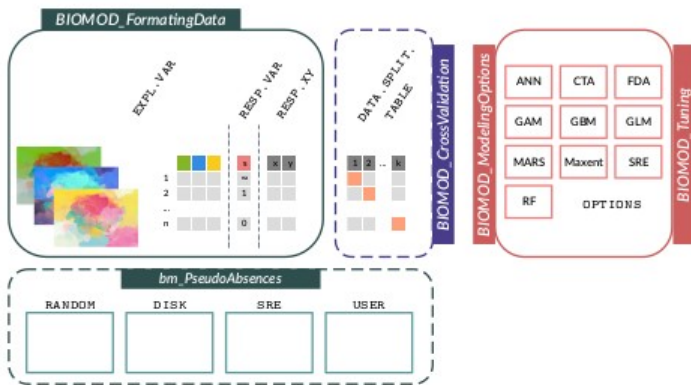
<https://rspatial.org/raster/sdm/>

- Profile methods
 - Bioclim
 - Domain
 - Mahalanobis distance
- Classical regression models
 - Generalized Linear Models
 - Generalized Additive Models
- Machine learning methods
 - Maxent
 - Boosted Regression Trees
 - Random Forest
 - Support Vector Machines

Bimod2 package

<https://biomodhub.github.io/biomod2/index.html>

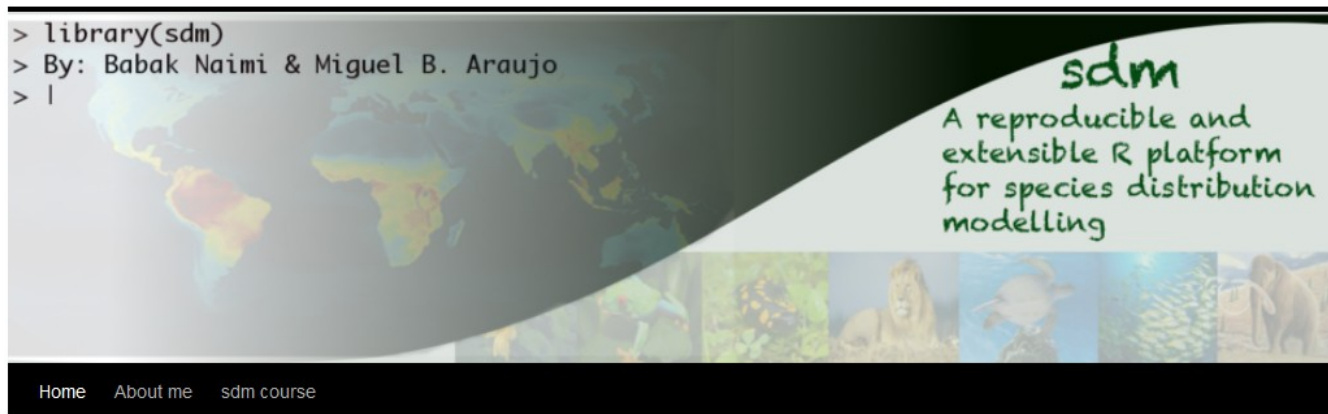
- Originate in 2009, the first ensemble model platform;
- Very powerful in post model processing;
- Most widely used climate change forecast;
- Powerful in multiple species/big data;
- Later, included Maxent, together with shinyapp;
- Developed by [Wilfried Thuiller](#), Damien Georges, [Maya Gueguen](#), Robin Engler, Frank Breiner, Bruno Lafourcade, Remi Patin.



Bimod2

sdm package

<https://www.biogeoinformatics.org;>



sdm

getting started with sdm package

[Babak Naimi](#), [Miguel B. Araujo](#)

3 April 2016 (update: 17 Oct. 2016)

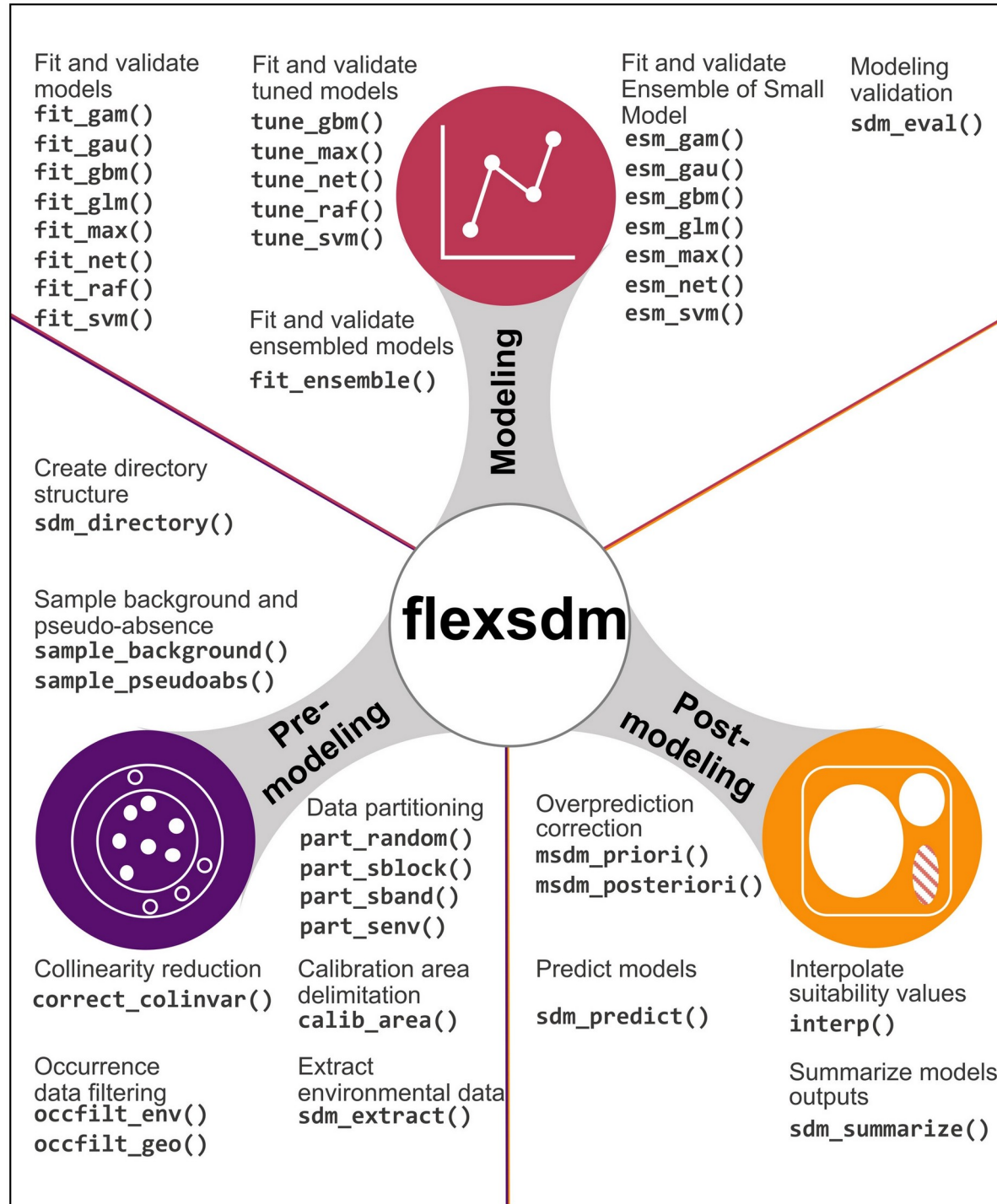
<https://sdm.r-forge.r-project.org/>

- Simplized version of biomod2;
- sdm is an object-oriented, reproducible and extensible, platform for species distribution modelling.
- Developed by Babak Naimi and **Miguel B. Araújo**

flexsdm package

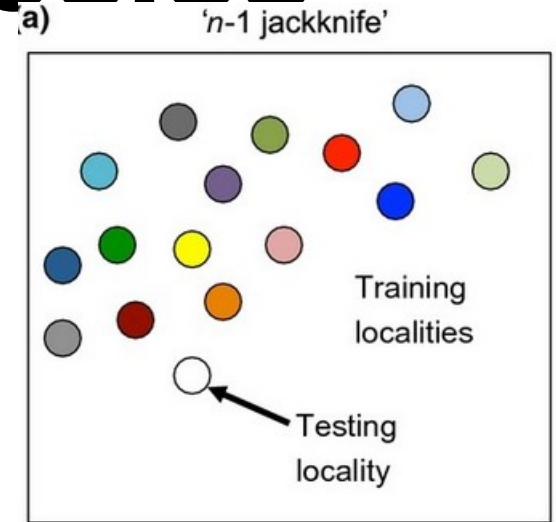
<https://sjevelazco.github.io/flexsdm/>

- Released last year 2022, still under development
- Developed by **Janet Franklin lab**, former chief editor of *Diversity and Distributions*
- Flexible:
 - pre-model,
 - modelling functions,
 - Post-modelling functions,
- Lack of function for summarizing model predictions;



Rare, endemic species

- Leaving-one-out method;
Proposed by Richard Pearson et al. 2007
lowest presence threshold (LPT)



leave-one-out (jackknife)

-
- **Ensemble of small model;**
Proposed by Guisan group
ecospat package

Building Ensembles of Small Models (ESMs)

Simple bivariate models

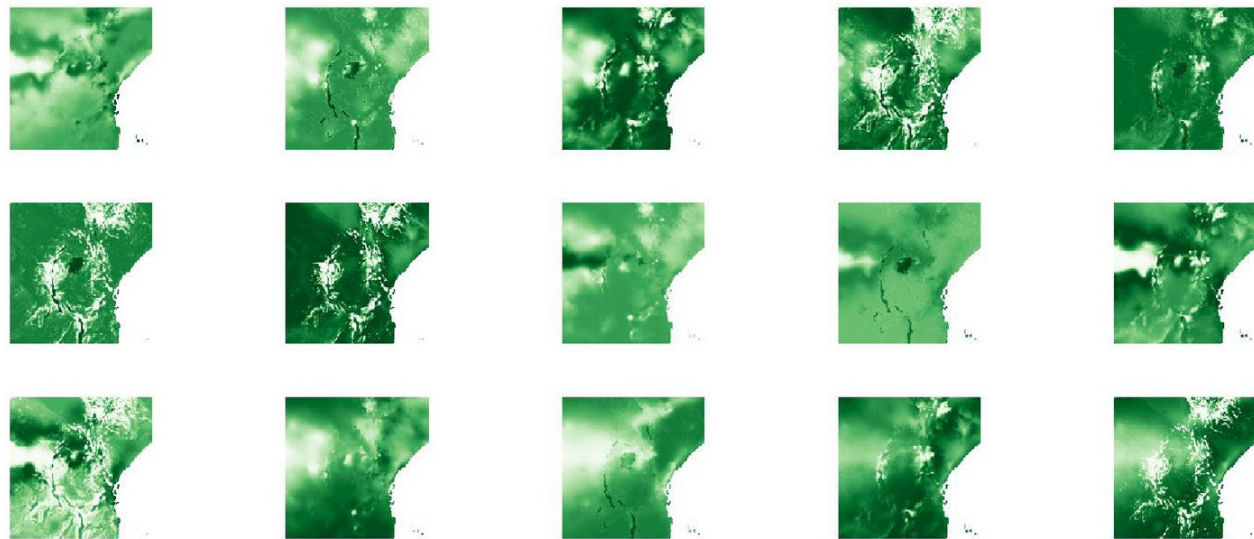
e.g.:

$biva_1: T_{mean} + P_{mean}$

$biva_2: T_{mean} + topo$

$biva_3: P_{mean} + topo$

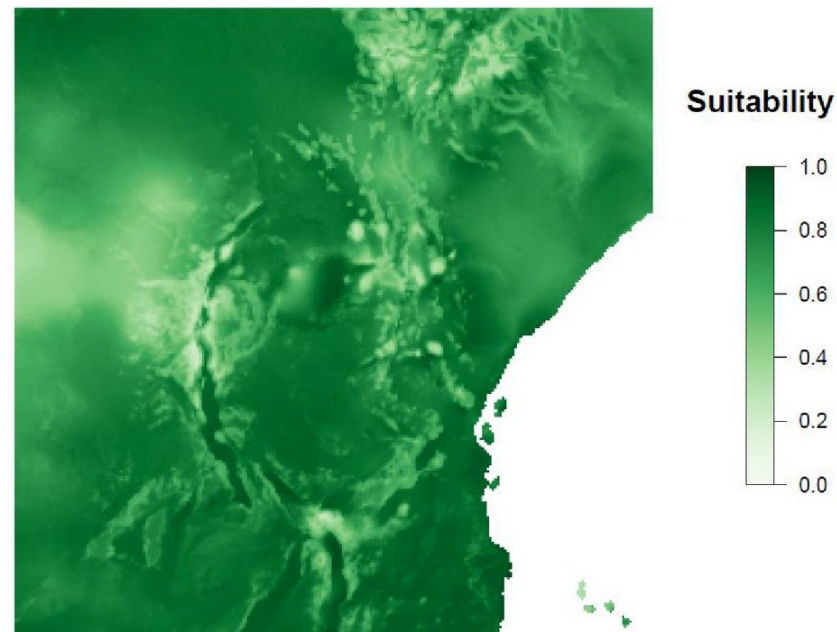
etc.



Ensemble of Small Models ESM

Average of small models weighted by a score from an evaluation index

$$ESM = \frac{\sum_{i=1}^n biva_i \times EI_i}{\sum_{i=1}^n EI_i}$$



Demo

https://sjevelazco.github.io/flexsdm/articles/v05_Rare_species_example.html

FORUM

Optimizing
distribution

Frank T. Breiner

Abstract

1. Ensembles of Small Models (ESM) represent a novel strategy for species distribution modelling with few observations. ESMs are built by calibrating many small models and then averaging them into an ensemble model where the small models are weighted by their cross-validated scores of predictive performance. In a previous paper (Breiner, Guisan, Bergamini, & Nobis, *Methods in Ecology and Evolution*, 6, 1210–1218, 2015), we reported two major findings. First, ESMs proved largely superior to standard models in terms of model performance and transferability. Second, ESMs including different modelling techniques did not clearly improve model performance compared to single-technique ESMs. However, ESMs often require a large computation effort, which can become problematic when modelling large numbers of species. Given the appealing new perspectives offered by ESMs, it is especially important to investigate if some techniques yield increased performance while saving computation time and thus could be predominantly used for building ESMs.
2. Here, we present results from a reanalysis of a subset of the data used in Breiner et al. (2015). More specifically, we ran ESMs: (1) fitted with 10 modelling techniques separately (in Breiner et al., 2015 we used only three techniques); and (2) using various parameter options for each modelling technique (i.e., model tuning).
3. We show that ESMs vary in model performance and computation time across techniques, and some techniques are advantageous in terms of optimizing model performance and computation time (i.e., GLM, CTA and ANN). Including one of these modelling techniques could thus optimize computation time compared to using more computing-intensive techniques like GBM. Next, we show that parameter tuning can improve performance and transferability of ESMs, but often at the cost of computation time. Parameter tuning could therefore be used when computing resources are not a limiting factor.
4. These findings help improve the applicability and performance of ESMs when applied to large numbers of species.